

Shi-Jie Gao (高世杰)

Postdoctoral Researcher

Compact-object Binaries & Radio Pulsars

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Employment and Education

- 2025.07–present **Postdoctoral Researcher**, *School of Astronomy and Space Science*, Nanjing University.
Current focus: radio pulsar and transient searches.
- 2020.09–2025.06 **Ph.D., Astronomy**, *School of Astronomy and Space Science*, Nanjing University.
Advisor: Prof. Xiang-Dong Li.
Doctoral focus: formation and evolution of compact-object binaries and X-ray binaries.
- 2016.09–2020.06 **Undergraduate Studies**, *School of Physics*, Shandong University.
Major: Physics.

Research Interests and Selected Projects

- Theory **Compact-object Formation and Binary Evolution.**
Compact-object binaries, X-ray binaries, binary pulsars, and mass-gap black holes.
Modeling their formation and evolution using stellar- and binary-evolution calculations.
- Observation **Radio Pulsars and Transients.**
Pulsar searches and polarization analysis with FAST, the Green Bank Telescope, and the Parkes 64-m radio telescope (Murriyang).
Highlights include the detection of more than 30 pulsars in FAST data, the discovery of the first pulsar associated with Terzan 6, and studies of RRAT J2325–0530.

Publications

First-author Publications

- 2026 **Gao, Shi-Jie**, Yi-Xuan Shao, and Xiang-Dong Li. Pulsar Gleaners: Discovery of 19 Pulsars in FAST Archival Data at $|b| < 5^\circ$ and Decl. $< -5^\circ$. *ApJ*, 997, 210, doi.
- 2026 **Gao, Shi-Jie**, Xiang-Dong Li, Song Wang, Kareem El-Badry, De-Jiang Zhou, Yi-Xuan Shao, Zhen Yan, Pei Wang, Ping Zhou, and Jin-Lin Han. Search for Radio Pulsations from Neutron Star Candidates in Detached Binaries. *ApJ*, 997, 222, doi.
- 2025 **Gao, Shi-Jie**, Xiang-Dong Li, Zhen Yan, Yi-Xuan Shao, and Ping Zhou. RRAT J2325–0530: A Rotating Radio Transient with an Atypical Waiting-time Distribution. *ApJ*, 991, 201, doi.
- 2025 **Gao, Shi-Jie**, Xiang-Dong Li, Yi-Xuan Shao, Ping Zhou, Pei Wang, Yun-Wei Yu, Zhen Yan, and Di Li. A Target Search for Fast Radio Bursts Associated with Two Fast Blue Optical Transients: AT2018cow and CSS161010. *ApJ*, 990, 93, doi.
- 2024 **Gao, Shi-Jie**, Yi-Xuan Shao, Pei Wang, Ping Zhou, Xiang-Dong Li, Lei Zhang, Joseph W. Kania, Duncan R. Lorimer, and Di Li. Discovery of a Millisecond Pulsar Associated with Terzan 6. *ApJ Letters*, 974, L2, doi.
- 2023 **Gao, Shi-Jie** and Xiang-Dong Li. The White Dwarf Mass–Orbital Period Relation under Wind Mass Loss. *MNRAS*, 525, 2605, doi.
- 2023 **Gao, Shi-Jie** and Xiang-Dong Li. Can Cosmologically Coupled Mass Growth of Black Holes Solve the Mass Gap Problem? *ApJ*, 956, 128, doi.

- 2022 **Gao, Shi-Jie**, Xiang-Dong Li, and Yong Shao. Formation of Mass-gap Black Holes from Neutron Star X-Ray Binaries with Super-Eddington Accretion. *MNRAS*, 514, 1054, doi.
- 2021 **Gao, Shi-Jie** and Xiang-Dong Li. On the Magnetic Fields of Ultraluminous X-Ray Pulsars. *Research in Astronomy and Astrophysics*, 21, 196, doi.

Coauthored Publications

- 2026 Yuan-Zhu Wang, Yin-Jie Li, **Gao, Shi-Jie**, Shao-Peng Tang, and Yi-Zhong Fan. A New Group of Low-spin $50\text{--}70 M_{\odot}$ Black Holes and the High Pair-instability Mass Cutoff. *Science China Physics, Mechanics & Astronomy*, 69, 299562, doi.
- 2026 Yu-Dong Nie, Yong Shao, Jian-Guo He, Ze-Lin Wei, **Gao, Shi-Jie**, Xiao-Jie Xu, and Xiang-Dong Li. Formation of Recycled Pulsars in Common Envelope Binaries. *ApJ*, 997, 265, doi.
- 2026 Yu-Feng Luo, Shan-Shan Weng, Qing-Zhong Liu, Ming-Yu Ge, Han-Long Peng, Shi-Qi Zhou, **Gao, Shi-Jie**, Yu-Jia Zheng, and Yan Zhang. The Orbital Parameters of Gamma-Ray Binary PSR J2032+4127. *ApJ*, 998, 332, doi.
- 2026 Hong-Bo Li, **Gao, Shi-Jie**, Xiang-Dong Li, and Ren-Xin Xu. Polar Mounds on Strangeon Stars: The Neutrino Emission from Ultraluminous X-Ray Pulsars. *MNRAS*, 546, ag241, doi.
- 2025 Yi-Ying Wang, **Gao, Shi-Jie**, and Yi-Zhong Fan. Probing Cosmology with 92 Localized Fast Radio Bursts and DESI BAO. *ApJ*, 981, 9, doi.
- 2025 Xiangyu Ivy Wang, **Gao, Shi-Jie**, and Xiang-Dong Li. A Two-stage Kick Scenario for the Peculiar Low-mass X-Ray Binary GX 1+4. *ApJ Letters*, 991, L37, doi.
- 2024 Hao-Ran Yang, Xiang-Dong Li, **Gao, Shi-Jie**, and Kun Xu. Instability in Supernova fallback Disks and Its Effect on the Formation of Ultralong Period Pulsars. *ApJ*, 976, 77, doi.
- 2024 Song Wang, Xinlin Zhao, Fabo Feng, Hongwei Ge, Yong Shao, Yingzhen Cui, **Gao, Shi-Jie**, et al. A Potential Mass-gap Black Hole in a Wide Binary with a Circular Orbit. *Nature Astronomy*, 8, 1583, doi.
- 2024 Yin-Jie Li, Shao-Peng Tang, **Gao, Shi-Jie**, Dao-Cheng Wu, and Yuan-Zhu Wang. Exploring Field-evolution and Dynamical-capture Coalescing Binary Black Holes in GWTC-3. *ApJ*, 977, 67, doi.
- 2023 Wen-Shi Tang, **Gao, Shi-Jie**, and Xiang-Dong Li. Impact of Asymmetrical Mass Ejection from Proto-white Dwarfs on the Properties of Binary Millisecond Pulsars. *MNRAS*, 519, 2951, doi.
- 2023 Jian-Guo He, Yong Shao, **Gao, Shi-Jie**, and Xiang-Dong Li. Detection Prospects of Fast-merging Gravitational Wave Sources in M31. *ApJ*, 953, 153, doi.

Selected Talks and Presentations

- 2024.12.16 **Discovery of a Millisecond Pulsar Associated with Terzan 6**, *The International Pulsar Symposium 2024*, Guiyang, China.
- 2024.10.11–10.12 **Discovery of a Millisecond Pulsar Associated with Terzan 6**, “Space-Time Probes” Pulsar Science Workshop, Huangshan, China.
- 2023.11.23 **The White Dwarf Mass-Orbital Period Relation under Wind Mass-Loss**, *Fifth Workshop on Multi-wavelength Studies of X-ray Binaries*, Wuhan, China.
- 2023.11.10 **Formation and Evolution of Binaries with Compact Objects**, *2023 Jiangsu Graduate Student Innovation and Practice Competition in Astronomy*, Nanjing, China.
- 2023.07.10 **The Neutron Star/Black Hole Mass Gap**, *Workshop on High-Energy Radiation from Compact Objects with Insight-HXMT*, Qingdao, China.
- 2023.04.28 **Formation of Mass-Gap Black Holes from X-ray Binaries with Super-Eddington Accretion**, *The 14th Zhang Heng Symposium*, Wuhan, China.
- 2021.12.25 **Low-Mass-Gap Black Holes from Neutron Stars with Super-/Sub-Eddington Accretion**, *Fourth Workshop on Multi-wavelength Studies of X-ray Binaries*, Online.

Teaching, Mentoring, and Public Outreach

NJU 4.5-m Radio Telescope (NJURT)

Ongoing Commissioning and testing at Nanjing University's Zuo Dijiang Observatory, including telescope control, pointing calibration, observing-chain tests, and development of an SDR-based pulsar observing terminal for teaching and public outreach; detected radio pulsations from eight pulsars, including the Crab and Vela pulsars and PSR B0329+54.

Student Mentoring

Current **Research Mentor**, *Can Xu and Yajing Ju, M.Sc. students*, Nanjing Normal University. Pulsar search projects.

Current **Research Mentor**, *Yihan Lu, undergraduate student*, Nanjing University. Pulsar observations with the NJU 4.5-m Radio Telescope at Zuo Dijiang Observatory.

Teaching Assistantships

2021–2025 **Teaching Assistant**, *A Brief History of the Universe (undergraduate course)*, Nanjing University, Instructor: Prof. Xiang-Dong Li. Nine semesters, from Spring 2021 through Spring 2025.

Fall 2021 **Teaching Assistant**, *Astrophysical Radiation Mechanisms (graduate course)*, Nanjing University.

Public Outreach

2026.04 **Invited Public Talk**, *From FAST to the 4.5-m Antenna: How Do We “Hear” Pulsars?*, NJU Astronomy Night VII.

2023.05.20 **Invited Public Talk**, *Stellar Evolution and Compact-Object Binaries*, School of Physics, Shandong University, Jinan, China.

2021 **Science Outreach Volunteer**, *Astronomy outreach events*, Nanjing University and Jiangsu Science and Technology Museum. Selected activities include NJU Astronomy Night II and a Science and Technology Week outreach event.

Training and Workshops

2026.05.15 **Parkes Remote Observer Training**, *Australia Telescope National Facility*, Virtual. Completed the training and became an approved remote observer for Murriyang, the Parkes radio telescope.

2024.02.06–02.08 **GBT Observer Training Workshop**, *Green Bank Observatory*, Virtual. Completed intensive Green Bank Telescope observing training and became a qualified GBT observer.

2025.08.01–08.10 **Pulsar Summer School**, *National Astronomical Observatories, Chinese Academy of Sciences*, Beijing, China.

2024.08.19–08.23 **ICESUN Summer School 2024**, *Yunnan Observatories, Chinese Academy of Sciences*, Kunming, China. Theme: *Stellar Explosions and Related Objects*. Training in stellar physics, binary evolution, massive-star evolution, common-envelope evolution, supernovae, and related observations and modeling.

2022.08.06–08.15 **Pulsar Summer School**, *National Astronomical Observatories, Chinese Academy of Sciences*, Xiamen, China.